

Magnet Magic — What Sticks?

Grade 2 Science Lesson Plan: Exploring Magnetism Through Hands-On Discovery

Subject	Science (Physical Science)
Grade Level	Grade 2
Duration	45–60 minutes
Topic	Magnets, magnetism, and magnetic vs. non-magnetic materials

Learning Objectives

- Explain that magnets have an invisible pulling force called magnetism.
- Identify materials that are attracted to magnets (iron, steel, nickel, cobalt) and materials that are not (wood, plastic, glass, fabric).
- Describe that magnets have two poles — North and South — and that opposite poles attract while same poles repel.
- Recognize common types of magnets, such as bar magnets and horseshoe magnets.
- Identify everyday uses of magnets, such as fridge magnets, compasses, and door catches.

Materials Needed

- Bar magnets and/or horseshoe magnets (one per group of 2–3 students)
- A collection of small objects to test: paperclip, plastic spoon, coin, pencil, glass marble, fabric scrap, metal nail, wooden block
- Student recording sheets (table for sticks / doesn't stick)
- Smiley face and thumbs-up stickers or stamps (optional, for recording results)
- A compass (optional, for the everyday-uses discussion)

Lesson Sequence

1. Introduction — What Is a Magnet? (5–8 minutes)

Begin by asking students if they have ever played with a magnet. Explain that a magnet is a special object with an invisible force that pulls certain metals — like iron and steel — toward it. Demonstrate by holding a magnet near a paperclip and letting students watch it "jump" toward the magnet.

2. How Do Magnets Work? (5 minutes)

Introduce the term **magnetism** — an invisible pulling force that acts like magic you cannot see. Emphasize two key ideas: magnetism is an invisible pulling force, and it pulls metals like iron and steel closer.

3. What Sticks and What Doesn't? (10 minutes)

Show students the two groups of materials below. Discuss each item and ask students to predict whether it will stick to a magnet before testing.

Sticks to a Magnet (Magnetic)	Does NOT Stick (Non-Magnetic)
Iron — nails and bolts	Wood — pencils and blocks
Steel — paperclips and cans	Plastic — toys and spoons
Nickel — found in some coins	Glass — marbles and jars
Cobalt — used in batteries	Fabric — clothes and bags

4. Types of Magnets (5 minutes)

Bar Magnet: A rectangular block with a North pole at one end and a South pole at the other — one of the most common magnets found in a science classroom.

Horseshoe Magnet: Bent into a U-shape so both poles face the same direction, making it extra strong and great for picking up metal objects.

5. Hands-On Experiment: What Sticks to Your Magnet? (10–15 minutes)

Guide students through the following steps in pairs or small groups:

- 1 Gather objects to test — a paperclip, plastic toy, coin, and pencil.
- 2 Touch the magnet to each object, one at a time.
- 3 Observe which objects stick and which do not.
- 4 Record results on the recording sheet using a smiley face (sticks) or thumbs-up (doesn't stick).

6. Safety Reminders

- Keep magnets away from electronics.
- Never put magnets near your mouth.
- Handle magnets gently to avoid pinching fingers.
- Store magnets safely when not in use.
- Ask an adult before using strong magnets.

7. Fun Magnet Facts (5 minutes)

- Magnets have two poles: North and South.
- Opposite poles attract, and same poles repel each other.
- Earth itself is a giant magnet with its own North and South poles!
- Some animals use Earth's magnetism to find their way home.
- Magnets can lose their power if they are heated too much.
- The strongest magnets are called superconducting magnets.

8. Magnets in Everyday Life (5 minutes)

Discuss how magnets are part of daily life: fridge magnets hold notes and drawings using magnetic force; compasses use a tiny magnet to point North; magnetic door catches keep cabinet doors shut; toys like magnetic tiles and electronics like speakers also rely on magnets. Ask students to look around their homes and try to spot a magnet.

9. Quiz Check (5 minutes)

Question: Which of these sticks to a magnet — a plastic spoon, a paperclip, a glass marble, or a wooden block?

Answer: The paperclip! It is made of metal, and metals like iron and steel are attracted to magnets.

Wrap-Up: What Did We Learn?

- Magnets pull iron and steel objects toward them.
- Magnets have two poles: North and South.
- Not all materials stick to a magnet.
- Opposite poles attract, same poles repel.
- Magnets are useful in everyday life.
- Earth itself is a giant magnet!

Assessment Ideas

Use the completed experiment recording sheets to assess whether students can correctly sort objects into magnetic and non-magnetic categories. Ask students to verbally explain, in their own words, why a paperclip sticks to a magnet but a plastic spoon does not.

Extension Activity

Have students go on a "magnet hunt" at home with an adult, searching for magnets in places like the refrigerator, cabinet doors, toy boxes, or backpacks, and report back on what they find.

Keep exploring, keep questioning, and keep discovering the magic around you!